

THEJUS MAHAJAN PH.D.

Computational Scientist · Clinical Data Pipelines · Biostatistics · Bioinformatics

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PROFILE

- Computational scientist (PhD, Post-doc) with formal further training in **applied bioinformatics and biostatistics**, and production experience in clinical data pipeline engineering.
- HealthTwiSt GmbH (Praxisphase): refactored a production R pipeline for the DeGIR quality registry (143,000+ patient records, ~300 clinics) with **byte-identical output verified at every step**; built the `degirtools` R package and a Shiny dashboard.
- Core skills: R (tidyverse, Shiny, tidymodels), Python (Pandas, NumPy, PyTorch), SQL, Git, Nextflow.
- Open to Clinical Data Science, Biostatistics, Statistical Programming, Clinical Data Management and Research Software Engineering; open to relocation within Germany. **Available immediately.**

PROFESSIONAL EXPERIENCE

Independent Research – Deep-Learning Pipeline Validation

Self-directed, open source

📅 07/2026 - present 📍 Hamburg, Germany

Correctness and validation of machine-learning analysis pipelines

- Diagnosed and corrected **two systematic errors** in an analysis pipeline that produced plausible but invalid results – neither raised an exception nor failed a test. Published the analysis, including a public correction of my own earlier findings.
- Built the toolchain (Python, **PyTorch**, ONNX) with an automated test suite whose guards are **mutation-tested**: each is verified to fail when the logic it protects is broken.
- Write-up: *The Mirror in the Machine* (thejusmahajan.github.io)

Clinical Data Engineering — Praxisphase (Weiterbildung ABI-2025-3)

HealthTwiSt GmbH

📅 02/2026 - 04/2026 📍 Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,349 lines (–26.4%) with byte-identical output verified via `identical()`, and 360 deprecated calls modernised to current tidyverse standards.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, so clinical domain experts and data stewards maintain the rules directly.
- Created the `degirtools` R package (9 exported functions, roxygen2 docs, devtools::`check()` passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io on a de-identified, perturbed extract rather than patient data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

Continuing Education – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

📅 08/2025 - 02/2026 📍 Berlin, Germany

Applied bioinformatics and biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.

KEY STRENGTHS

Clinical Data Pipeline Engineering

Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.4% code reduction. Built `degirtools` R package. Byte-level verification after every change via `identical()`.

Verification & Correctness

Byte-identical verification of every pipeline change (`identical()`); mutation-tested guards; two silent correctness errors found in my own analysis code, fixed and published. Two pre-existing bugs found in production code and reported, not silently changed.

Biostatistics, Bioinformatics & Reproducible Research

Multivariate analysis (PCA, ANOVA), clinical outcome analysis, tidymodels workflows and cross-validation; NGS analysis, reproducible Nextflow (DSL2) pipelines, GDPR-compliant data handling.

TECHNICAL SKILLS

Programming

R (tidyverse, Shiny, tidymodels), Python (Pandas, NumPy, PyTorch, JAX), SQL / MySQL, Fortran, BASH

Clinical Data & Biostatistics

Medical registry pipelines, GDPR/DSGVO compliance, multivariate analysis, PCA, ANOVA, survival analysis concepts, clinical endpoints

Data Engineering

Pipeline refactoring, config-driven ETL, `identical()` verification, R Shiny dashboards, large datasets (HDF5, NetCDF)

Bioinformatics

NGS analysis, Nextflow (DSL2), sequence analysis

Machine Learning

tidymodels and scikit-learn workflows, cross-validation, PyTorch, transformer models, ONNX conversion

Tools

Git/GitHub, Linux/Unix, Linux clusters, Conda, LaTeX, Docker concepts

LANGUAGES

English

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Fluent (C1)

German

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B1 (Goethe-Zertifikat), B2 in preparation

- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Next-Generation Sequencing; Nextflow); structural bioinformatics.
- Biostatistics (R): ANOVA, PCA, HCA, multivariate analysis, Markov chains.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (🔗 GitHub).

Guest Scientist (parallel to continuing education from 08/2025) Helmholtz-Zentrum Hereon, Ecosystem Modelling

📅 05/2025 - 10/2025 📍 Geesthacht, Germany

Modelling evolutionary processes in marine ecosystems; simulation of long-term environmental impacts using advanced computational methods. Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Course

📅 02/2025 - 04/2025 📍 Hamburg, Germany

Active job search and German language acquisition (Goethe-Zertifikat B1).

Post-doctoral Scientist

Universität Hamburg, Institute for Marine Ecosystem and Fishery Science

📅 08/2021 - 01/2025 📍 Hamburg, Germany

Marine ecosystem modelling

- *Parental leave from 01/2024 to 12/2024.*
- Model development: Developed the “Cyanobacteria Life Cycle” (CLC) model in two formulations: 0-D agent-based, and 1-D concentration-based (advection–diffusion) within **GOTM-FABM**, for climate-warming projections.
- Ported a legacy Fortran/OpenMP engine to **Google JAX**, rewriting per-agent branching into vectorised, branchless form for GPU/TPU portability; run on Google Colab (T4 GPU). Large NetCDF datasets on a **Linux cluster**.

Career Transition

📅 04/2021 - 07/2021 📍 Hamburg, Germany

Relocation from India to Germany and active job search.

Physics Tutor & Chess Coach — Freelance / Tutorwaves Solutions Inc., Kerala, India · 10/2018 – 03/2021

Doctoral Researcher in Astrochemistry

Université Paris-Saclay

📅 10/2015 - 09/2018 📍 Orsay, France

Atomic collision experiments, data analysis and modelling on the AGAT setup at the Orsay Tandem accelerator.

- Analysed multi-detector coincidence data from three accelerator campaigns in the DP2 framework, working from NDB-formatted detector output.
- Diagnosed **incomplete-record rates of 15–50%**, traced them to detector positioning, corrected them, and validated the correction by repeating the campaign with a different detector configuration.
- Mass calibration against a tri-alpha source (Pu-239, Am-241, Cm-244; linear fit, $R^2 = 0.99$); timing-based noise rejection; background subtraction by paired control runs; uncertainties quantified per campaign.
- Adapted existing one- and two-electron **Classical Trajectory Monte Carlo** codes to the collision system and ran 10^6 -trajectory simulations with a documented convergence study, at Universidad Autónoma de Madrid and CELIA, Bordeaux.
- Contributed measured dissociation branching ratios and cross sections to the international **KIDA** astrochemical database, for reuse in other groups' chemical models.

French



Basic knowledge

Malayalam



Native

Tamil, Hindi



Working proficiency

CERTIFICATIONS & TRAINING

Deployable Data Analysis & AI Pipelines with HPC

Supercomputing-Akademie, HLRS

Universität Stuttgart | 09–10/2026 (current)

IBM AI Engineering Professional Certificate

Coursera

Deep Learning with PyTorch - completed 08/2026, credential DDDI9T0KHUJ4

Machine Learning with Python - in progress, expected 09/2026

Applied Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH, Berlin

advanced vocational training, completed 04/2026

Introduction to Supercomputing at JSC

Jülich Supercomputing Centre,
Forschungszentrum Jülich | 11/2025

PUBLICATIONS

Four peer-reviewed journal articles and two refereed conference proceedings in atomic and molecular physics / astrochemistry (2017–2020); a first manuscript from the marine ecosystem modelling post-doc is in preparation.

🔖 Google Scholar

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO)

10/2015 - 09/2018 Orsay, France

Supervisor: Dr. Karine Béroff.

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 12/2014 Calicut, India

First Class with Distinction (CGPA 8.71/10).

Hamburg, 8 September 2026

Research Project – Theoretical Atomic Physics

IIT Mandi (under Dr. Hari Varma)

02/2015 - 08/2015 Mandi, India

Theoretical atomic physics; partly remote.

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012 Thrissur, India

Grade: B+ (CGPA 3.49/4.0).



Dr. Thejus Mahajan